



THE B.E.S.T. STANDARDS

Benchmarks for Excellent Student Thinking

Algebra 1 EOC Assessment

Spring 2024 Released Test Items

B.E.S.T. Algebra 1 EOC Mathematics Reference Sheet

Customary Conversions

1 foot = 12 inches
1 yard = 3 feet
1 mile = 5,280 feet
1 mile = 1,760 yards

1 cup = 8 fluid ounces
1 pint = 2 cups
1 quart = 2 pints
1 gallon = 4 quarts

1 pound = 16 ounces
1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters
1 meter = 1000 millimeters
1 kilometer = 1000 meters
1 liter = 1000 milliliters

1 gram = 1000 milligrams
1 kilogram = 1000 grams

Time Conversions

1 minute = 60 seconds
1 hour = 60 minutes
1 day = 24 hours
1 year = 365 days
1 year = 52 weeks

Formulas

Forms of Linear Equations	Forms of Quadratic Functions	Forms of Exponential Functions
$y = mx + b$ $Ax + By = C$ $y - y_1 = m(x - x_1)$	$f(x) = ax^2 + bx + c$ $f(x) = a(x - h)^2 + k$ $f(x) = a(x - p)(x - q)$	$f(x) = ab^x$ $f(x) = a(1 \pm r)^x$

Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

where $ax^2 + bx + c = 0$ and $a \neq 0$

Final Amounts under Simple Interest	Final Amounts under Compound Interest
$A = P(1 + rt)$ where P = principal, r = rate, and t = time	$A = P\left(1 + \frac{r}{n}\right)^{nt}$ where P = principal, r = rate, n = number of times compounded, and t = time

B.E.S.T. Algebra 1 EOC

1. In baseball, a pitcher's earned run average is used to measure how effective the pitcher is. The earned run average, A , is calculated using the formula shown, where E is the number of earned runs and N is the number of innings the pitcher pitched.

$$A = \frac{9E}{N}$$

Which equation correctly solves the formula for the variable E ?

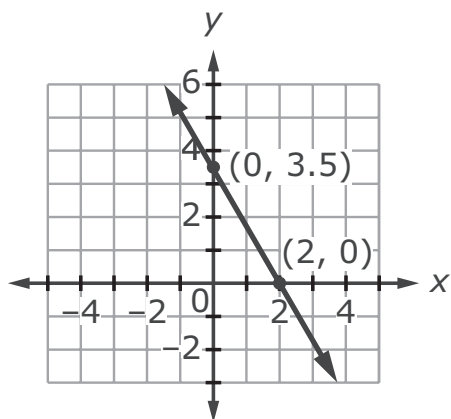
- Ⓐ $E = \frac{A}{9N}$
- Ⓑ $E = \frac{AN}{9}$
- Ⓒ $E = \frac{9A}{N}$
- Ⓓ $E = 9AN$

2. A function is shown.

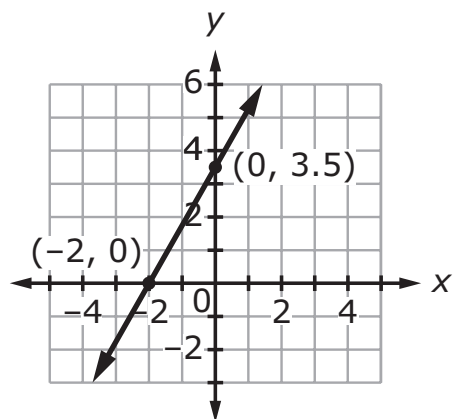
$$7x + 4y = 14$$

Which graph represents the function?

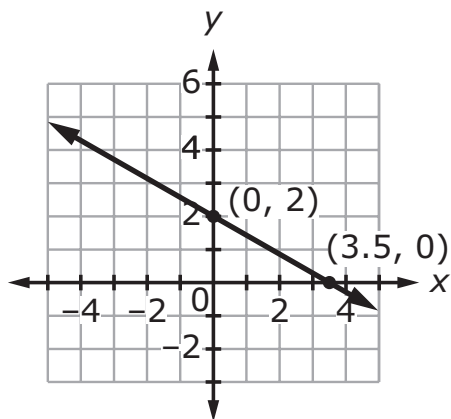
Ⓐ



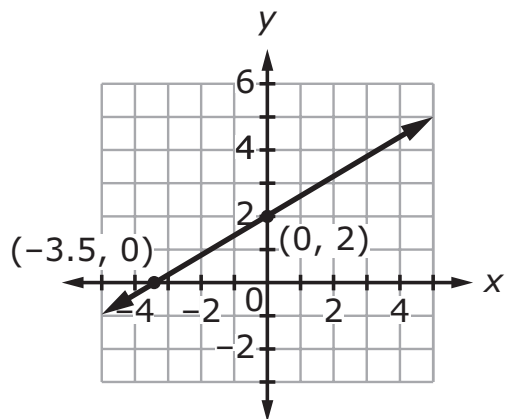
Ⓒ



Ⓑ



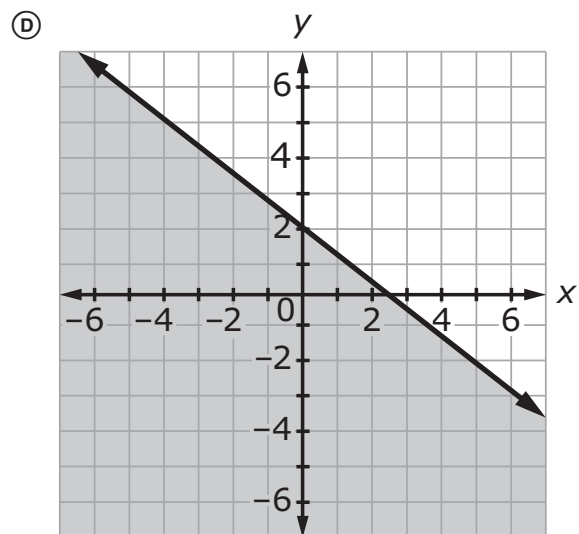
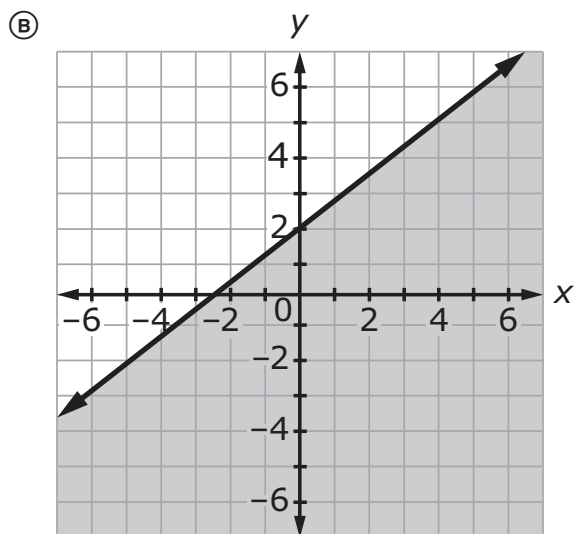
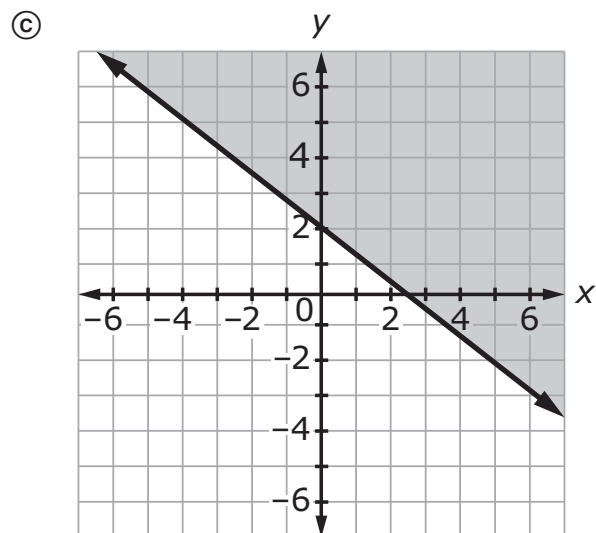
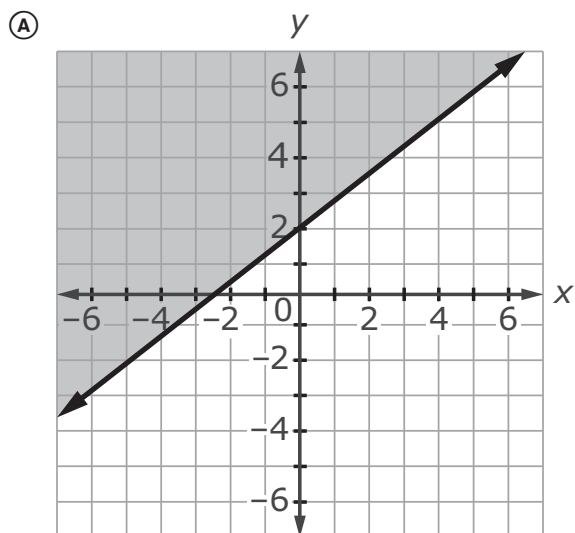
Ⓓ



3. An inequality is shown.

$$4x + 5y \geq 10$$

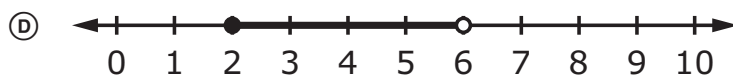
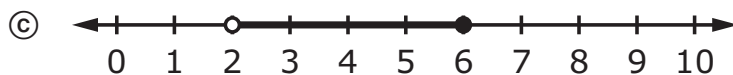
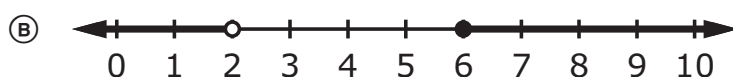
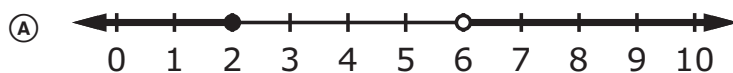
Which graph represents the solution set of the inequality?



4. A compound inequality is shown.

$$2x - 1 \leq 3 \text{ or } \frac{1}{2}x + 4 > 7$$

Which graph represents the solution set of the compound inequality?



5. An inequality is shown.

$$-3x + 7 \geq -x - 5$$

What is the solution set for x ?

- Ⓐ $\{x \mid x \leq 6\}$
- Ⓑ $\{x \mid x \geq 6\}$
- Ⓒ $\{x \mid x \leq -6\}$
- Ⓓ $\{x \mid x \geq -6\}$

6. What is the sum of $(2x^3 + 4x)$ and $(3x^3 + 5x)$?

- Ⓐ $14x^6$
- Ⓑ $14x^8$
- Ⓒ $5x^3 + 9x$
- Ⓓ $5x^6 + 9x^2$

7. A system of equations is shown.

$$\begin{aligned}y &= 2x + 2 \\ 2x + 5y &= 19\end{aligned}$$

What is the solution to the system?

Ⓐ $\left(\frac{3}{4}, 3\frac{1}{2}\right)$

Ⓑ $\left(1\frac{1}{8}, 4\frac{1}{4}\right)$

Ⓒ $\left(1\frac{2}{3}, 5\frac{1}{3}\right)$

Ⓓ $\left(3\frac{1}{4}, 2\frac{1}{2}\right)$

8. Which expression is equivalent to $\sqrt{3}$?

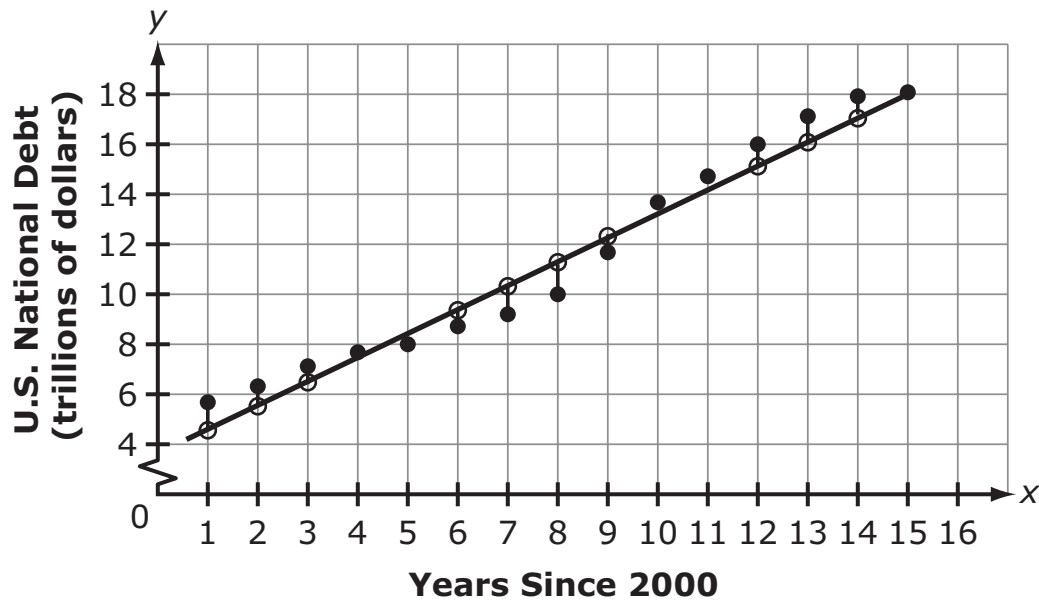
Ⓐ $2^{\frac{1}{3}}$

Ⓑ $3^{\frac{1}{2}}$

Ⓒ $\left(\frac{1}{2}\right)^3$

Ⓓ $\left(\frac{1}{3}\right)^2$

9. A scatter plot of the U.S. national debt from 2001 to 2016, including a linear equation and its residuals modeling the data, is shown.



Which statement correctly interprets the scatter plot?

- Ⓐ The scatter plot shows a weak positive correlation.
- Ⓑ The scatter plot shows a weak negative correlation.
- Ⓒ The scatter plot shows a strong positive correlation.
- Ⓓ The scatter plot shows a strong negative correlation.

10. An equation is shown.

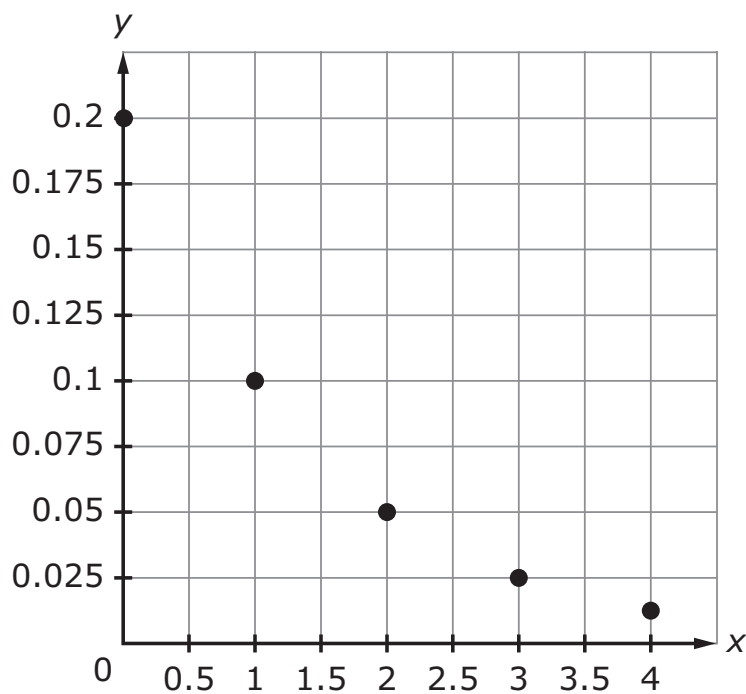
$$3x^2 - 21 = 222$$

What is one exact solution to the equation?

$x =$

← → ↶ ↷ ✖											
1	2	3	+	-	•	÷					
4	5	6	<	≤	=	≥	>				
7	8	9	$\square^\square$	$\square_\square$	()		$\sqrt{\square}$	$\sqrt[\square]{\square}$	π	i	
	0		sin	cos	tan	arcsin	arccos	arctan			
.	-	$\frac{\square}{\square}$									

- 11.** Five points from a function, f , are plotted in the first quadrant of the coordinate plane, as shown.



Which equation could represent the function?

- Ⓐ $f(x) = 0.2(1 - 0.1)^x$
- Ⓑ $f(x) = 0.2(1 - 0.5)^x$
- Ⓒ $f(x) = 0.1(1 - 0.05)^x$
- Ⓓ $f(x) = 0.1(1 - 0.25)^x$

12. Which expression is equivalent to $(7k + 2)(5k - 9)$?

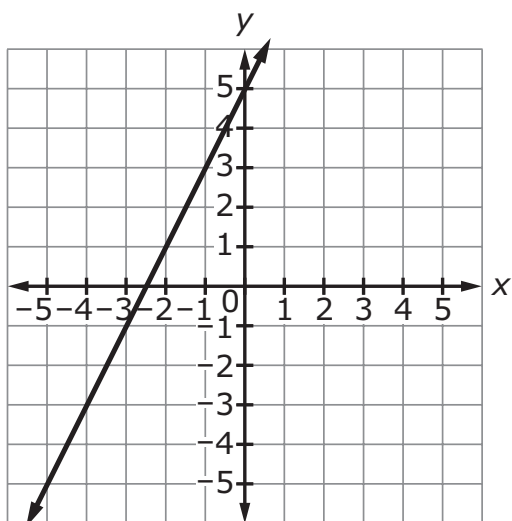
- Ⓐ $35k^2 - 18$
- Ⓑ $-18k - 18$
- Ⓒ $35k^2 - 53k - 18$
- Ⓓ $35k^2 - 73k - 18$

13. A function is shown.

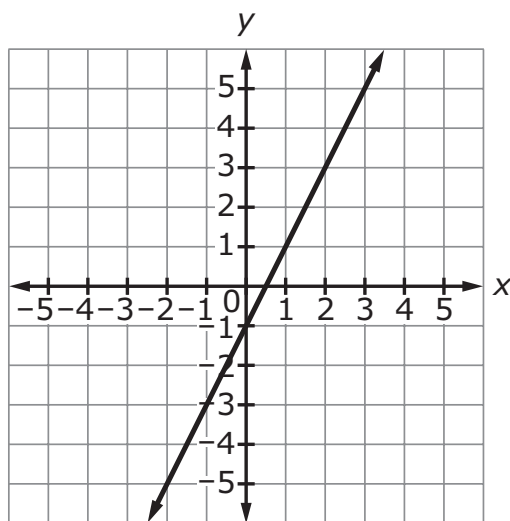
$$y + 1 = 2(x + 3)$$

Which graph represents the function?

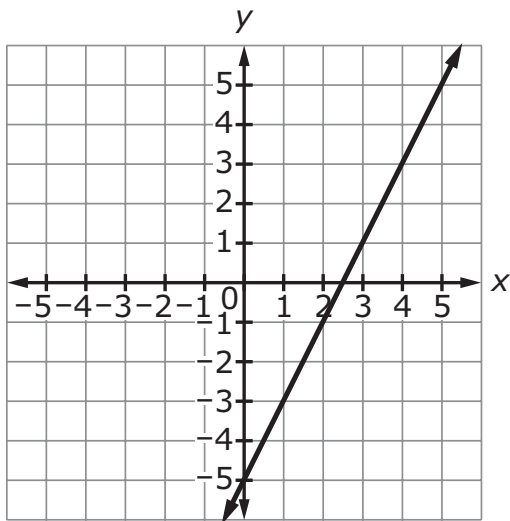
Ⓐ



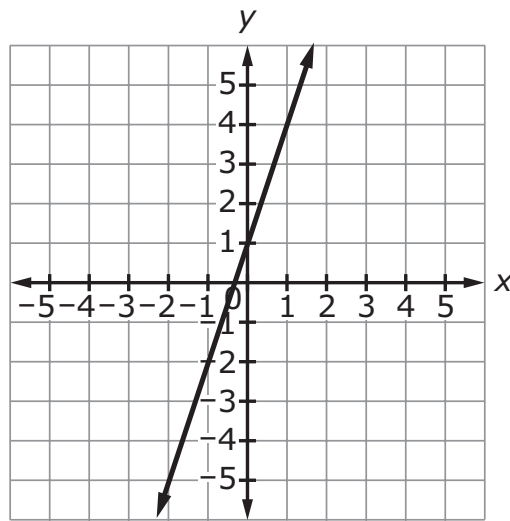
Ⓒ



Ⓑ



Ⓓ



- 14.** Garrett buys a large bag of dog food. He feeds his dog 2 cups of dog food each day.

Which statement describes the type of function that represents the amount of dog food in the bag after x days?

- Ⓐ positive linear
- Ⓑ negative linear
- Ⓒ exponential
- Ⓓ quadratic

15. Which table could represent a quadratic function?

Ⓐ

x	$f(x)$
0	3
1	9
2	27
3	81

Ⓒ

x	$h(x)$
-1	6
0	5
1	6
2	9

Ⓑ

x	$g(x)$
0	-2
1	-4
2	-6
3	-8

Ⓓ

x	$j(x)$
-3	7
-2	7
-1	7
0	7

16. Which function has the greatest slope?

Ⓐ $y = \frac{5}{3}x + 1$

Ⓑ $y = -x + 8$

Ⓒ $y = \frac{3}{4}x - \frac{5}{4}$

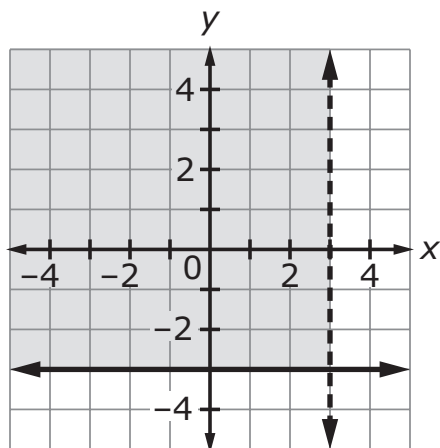
Ⓓ $y = \frac{3}{2}x + 3$

17. A system of inequalities is shown.

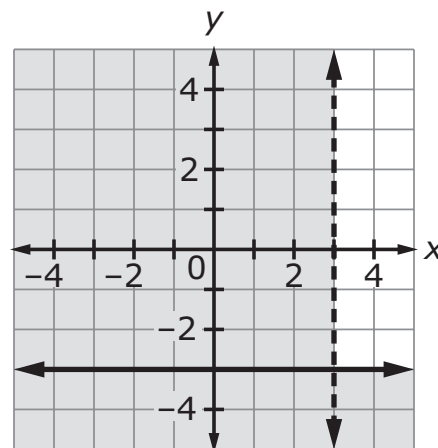
$$\begin{aligned} x &< 3 \\ y &\geq -3 \end{aligned}$$

Which graph represents the solution set?

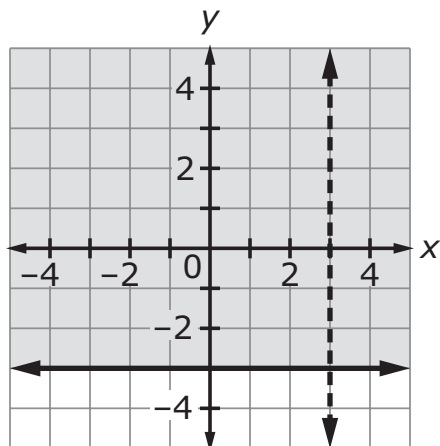
Ⓐ



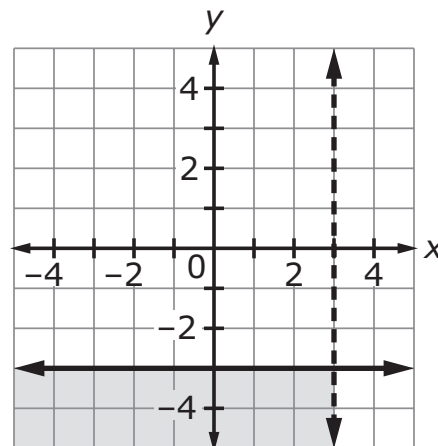
Ⓒ



Ⓑ



Ⓓ



- 18.** A ball was dropped from a balcony at a height of 20 feet above level ground. The height, in feet, of the ball after t seconds is modeled by the function shown.

$$y = -16t^2 + 20$$

What is the range of the function in terms of the situation described?

- Ⓐ $0 \leq t$
- Ⓑ $0 \leq y$
- Ⓒ $0 \leq t \leq 20$
- Ⓓ $0 \leq y \leq 20$

19. Which equation models exponential growth?

Ⓐ $y = 2\left(\frac{1}{5}\right)^x$

Ⓑ $y = 2(5.0)^x$

Ⓒ $y = 2\left(1 - \frac{1}{5}\right)^x$

Ⓓ $y = 2(1 - 0.5)^x$

20. Which function decays exponentially?

Ⓐ $f(x) = \frac{1}{5}x$

Ⓑ $g(x) = 5x$

Ⓒ $h(x) = 5^x$

Ⓓ $j(x) = \left(\frac{1}{5}\right)^x$

21. A line with a slope of -0.2 passes through the point $(11, 5)$.

Select numbers to complete the equation of the line in point-slope form.

$$y - [\textcircled{A} -0.2 \textcircled{B} 0 \textcircled{C} 5 \textcircled{D} 11] = [\textcircled{A} -0.2 \textcircled{B} 0 \textcircled{C} 5 \textcircled{D} 11]$$
$$(x - [\textcircled{A} -0.2 \textcircled{B} 0 \textcircled{C} 5 \textcircled{D} 11])$$

22. An expression is given, where $x \neq 0$.

$$\frac{4x^2 - 18x + 6}{2x}$$

Which expression is equivalent?

Ⓐ $2x - 9 + \frac{3}{x}$

Ⓑ $2x^2 - 16x + 4$

Ⓒ $2x^3 - 9x^2 + 3x$

Ⓓ $\frac{2}{x^3} - \frac{9}{x^2} + \frac{3}{x}$

23. An equation is given.

$$y = (x - 3)^2 + 1$$

Which type of function does the equation represent?

- Ⓐ cubic
- Ⓑ quadratic
- Ⓒ exponential
- Ⓓ square root

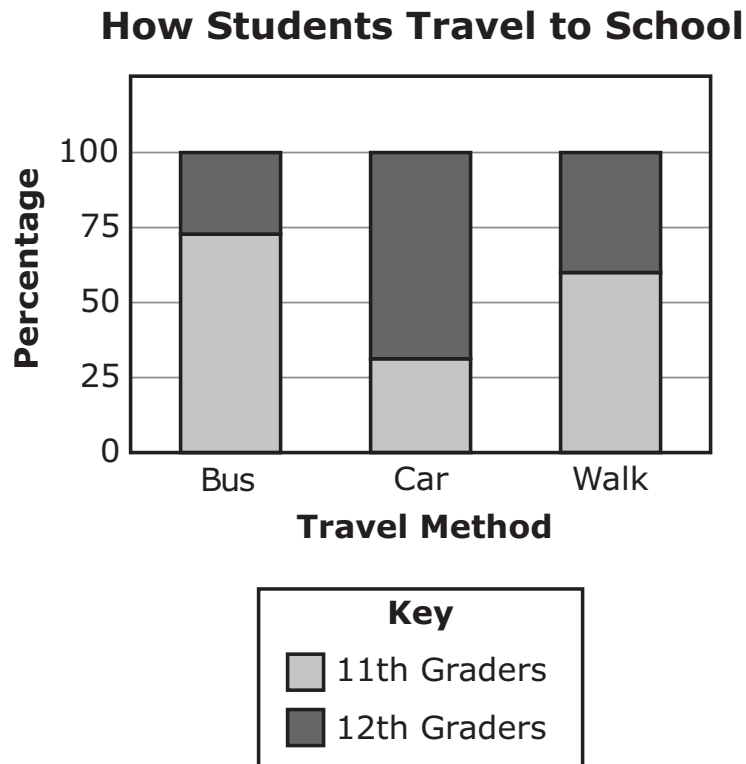
24. A function is given

$$h(x) = \frac{1}{5}x^2$$

How can be $h(15)$ evaluated?

- Ⓐ by multiplying $\frac{1}{5}$ by the square of 15
- Ⓑ by multiplying the square of $\frac{1}{5}$ by the square of 15
- Ⓒ by multiplying $\frac{1}{5}$ by the square of 15, then dividing by 15
- Ⓓ by multiplying the square of $\frac{1}{5}$ by the square of 15, then dividing by 15

25. Maribel asked a random group of upper-class students at her school which grade they are in and how they traveled to school in the morning. The data are shown in the segmented bar chart.



Which pair of terms describes the data?

- Ⓐ bivariate and numerical
- Ⓑ bivariate and categorical
- Ⓒ univariate and numerical
- Ⓓ univariate and categorical

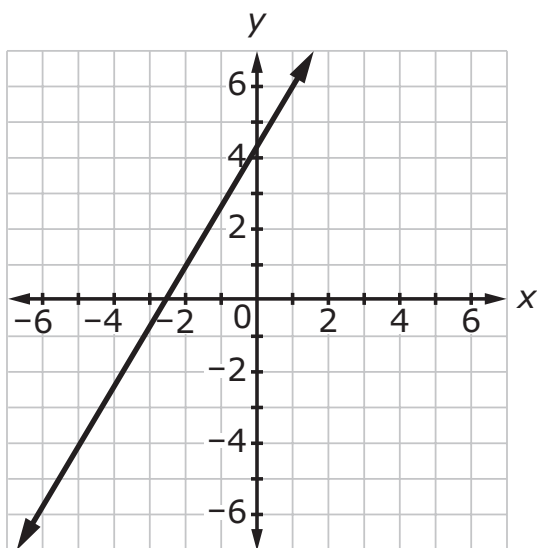
26. An expression is given.

$$42r^3s^2 - 35r^2s^3$$

Which expression is equivalent?

- Ⓐ $7rs(6s - 5r)$
- Ⓑ $7rs(6rs - 5rs)$
- Ⓒ $7r^2s^2(6r - 5s)$
- Ⓓ $7r^2s^2(6s - 5r)$

27. The graph of a line is shown.



Which equation represents the line?

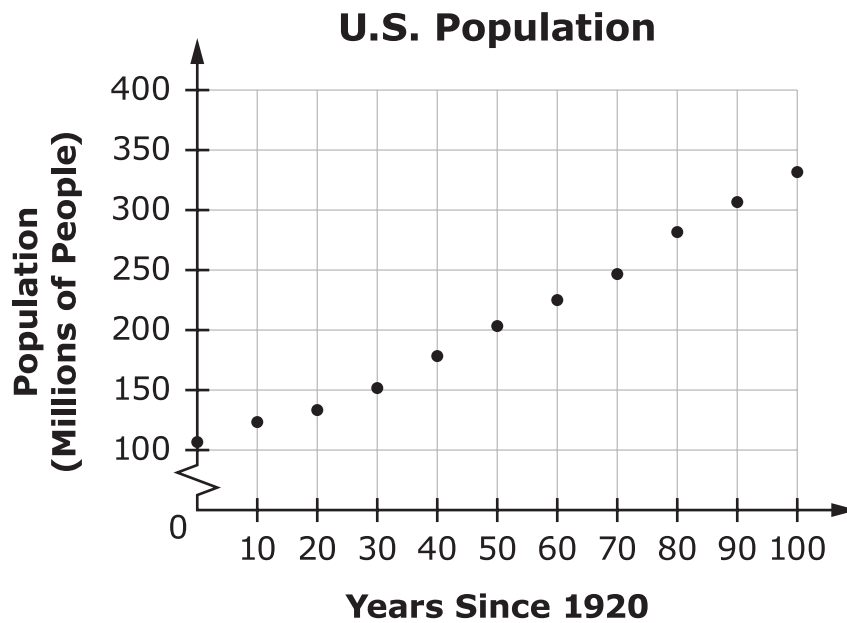
Ⓐ $y = \frac{3}{5}x - \frac{13}{5}$

Ⓑ $y = \frac{3}{5}x + \frac{13}{3}$

Ⓒ $y = \frac{5}{3}x - \frac{13}{5}$

Ⓓ $y = \frac{5}{3}x + \frac{13}{3}$

28. Darren collects data on the U.S. population from 1920 to 2020. He graphs the population and the number of years since 1920 in the scatter plot shown.



He determines the equation for the line of best fit to be $y = 2.33x + 92.05$ where y represents the population, in millions of people, and x represents the number of years since 1920.

Based on Darren's model, what is the projected U.S. population, in millions of people, for 2040?

←
→
↶
↷
✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

B.E.S.T. Algebra 1 EOC

29. Bashir has \$3410 to invest in two different investment accounts, Account A and Account B. The accounts earn simple interest for one year.

- Account A earns 6% interest.
- Account B earns 2% interest.
- He wants to earn at least \$196 in interest.

Bashir puts x dollars in Account A and y dollars in Account B.

Which system of constraints represents the situation?

- Ⓐ $x + y = 3410$
 $0.02x + 0.06y \geq 196$
- Ⓑ $x + y = 3410$
 $0.06x + 0.02y \geq 196$
- Ⓒ $0.02x + 0.06y = 3410$
 $x + y \geq 196$
- Ⓓ $0.06x + 0.02y \geq 3410$
 $x + y \geq 196$

- 30.** A rental company charges \$2 per hour to rent a bicycle. A table showing the prices of renting one bicycle for different amounts of time is shown.

Time (hours)	Price (dollars)
1	2
2	4
3	6

Which type of function models the price for renting one bicycle as a function of time?

- Ⓐ increasing linear
- Ⓑ decreasing linear
- Ⓒ increasing exponential
- Ⓓ decreasing exponential

- 31.** An artist earns a profit from selling y paintings. His profit, P , in dollars, is modeled with the given equation.

$$P = (200y + 23.25) - (10.53y)$$

His profit is the difference in the amount of money he earns from selling his paintings and the cost of making the paintings.

Which expression represents the cost of making y paintings?

- Ⓐ 23.25
- Ⓑ $10.53y$
- Ⓒ $200y + 23.25$
- Ⓓ $189.47y + 23.25$

- 32.** Cardiac output, D , is the amount of blood that the heart pumps per minute. The value of D can be found using the formula shown, where V is the amount of blood that the heart pumps with each heartbeat and R is the number of heartbeats per minute.

$$D = V \cdot R$$

Complete the equation to represent V in terms of D and R .

$V =$

← → ↶ ↷ ✖											
1	2	3	D	R							
4	5	6	+	-	•	÷					
7	8	9	<	≤	=	≥	>				
	0		$\square^\square$	$\square_\square$	()		$\sqrt{\square}$	$\sqrt[\square]{\square}$	π	i	
.	-	$\frac{\square}{\square}$	sin	cos	tan	arcsin	arccos	arctan			

B.E.S.T. Algebra 1 EOC

- 33.** Two divers are underwater. One diver is at a depth of 32 meters, while the other diver is at a different depth. The given equation can be used to find the possible depths, x , in meters, of the other diver.

$$|x-32| = 11$$

Enter values to complete the sentence about the other diver's depth, in meters.

The other diver is at a depth of either

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

or

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

meters.

34. The equation of a linear function, q , and a table of values of a linear function, r , are shown.

$$q(x) = 2x - 1$$

x	$r(x)$
0	-2
1	4
2	10
3	16

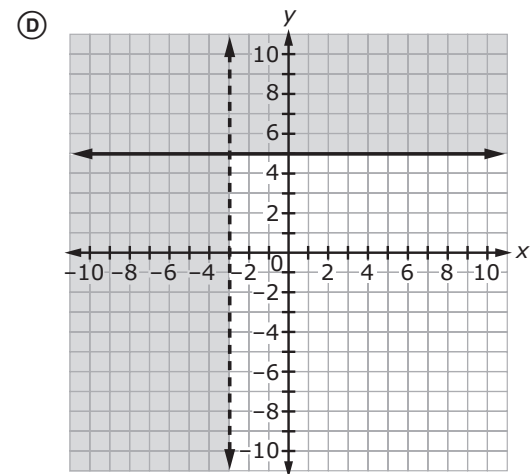
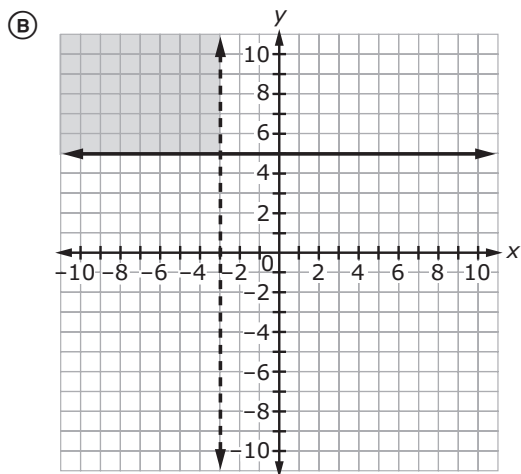
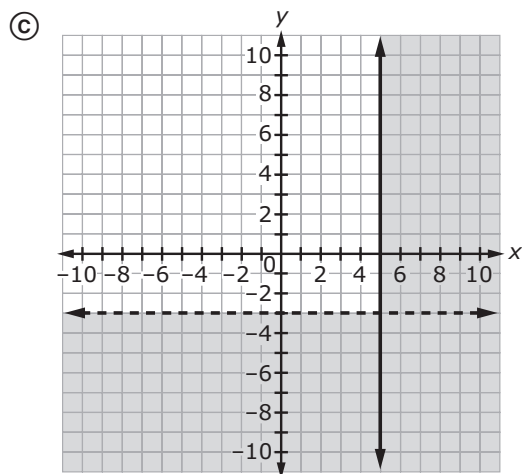
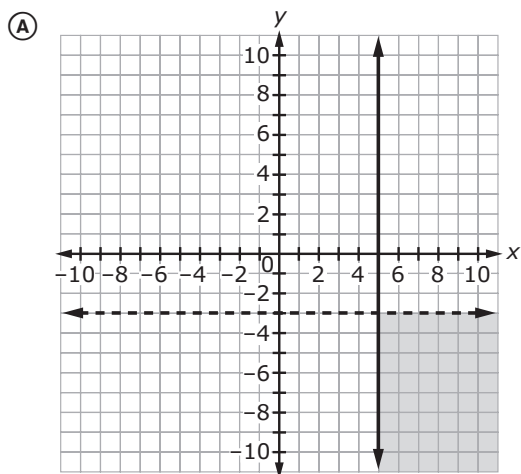
Which statement correctly compares the slopes and y -intercepts of the two functions?

- Ⓐ Function r has a lesser y -intercept and a slope that is 2 times the slope of function q .
- Ⓑ Function r has a lesser y -intercept and a slope that is 3 times the slope of function q .
- Ⓒ Function r has a greater y -intercept and a slope that is 2 times the slope of function q .
- Ⓓ Function r has a greater y -intercept and a slope that is 3 times the slope of function q .

35. A system of inequalities is given.

$$\begin{aligned} x &\geq 5 \\ y &< -3 \end{aligned}$$

Which graph represents the solution set?



36. An equation is shown.

$$x^2 - 15x = -14$$

What are the two solutions to the equation?

$x =$

← → ↶ ↷ ✖									
1	2	3							
4	5	6							
7	8	9							
	0								
.	-	$\frac{\Box}{\Box}$							

$x =$

← → ↶ ↷ ✖									
1	2	3							
4	5	6							
7	8	9							
	0								
.	-	$\frac{\Box}{\Box}$							

37. A function, f , is given.

$$f(x) = x^2 + 12x + 20$$

Select terms to describe the end behavior of the function.

As x approaches $-\infty$, $f(x)$ approaches [☐ A $-\infty$ ☐ B ∞ ☐ C 0] .

As x approaches ∞ , $f(x)$ approaches [☐ A $-\infty$ ☐ B ∞ ☐ C 0] .

38. The expression $\sqrt[3]{18} \cdot 5\sqrt[3]{6}$ is equivalent to $15\sqrt[3]{x}$.

What is the value of x ?

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

- 39.** Margo surveys 150 high school juniors and seniors about their favorite types of dipping sauce. She records the results in the table shown.

Favorite Dipping Sauce

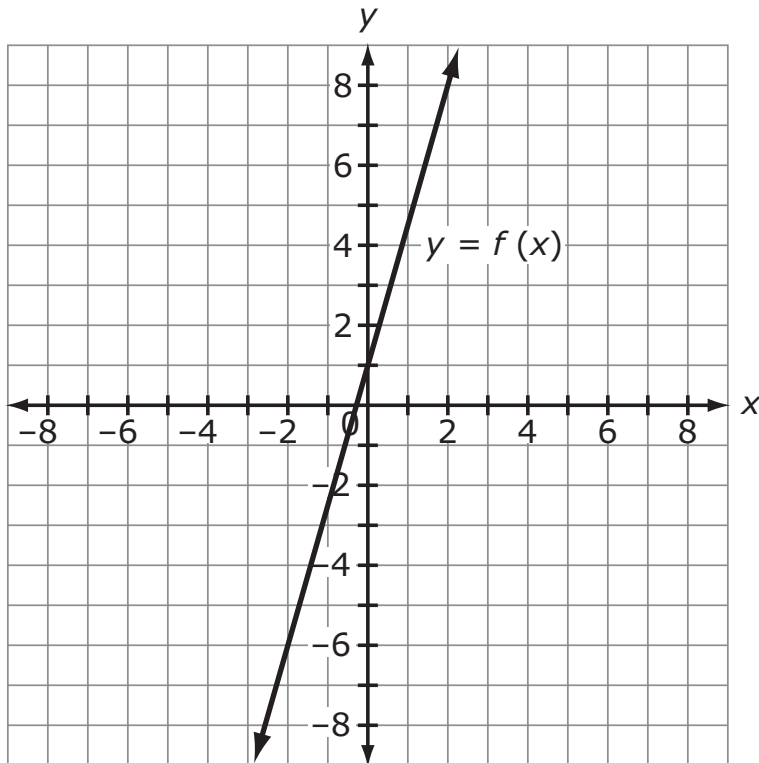
	Barbeque	Honey Mustard	Ranch	Sweet and Sour
Juniors	19	26	23	11
Seniors	7	22	30	12

How many more high school juniors and seniors chose honey mustard sauce than sweet and sour sauce?

←
→
↶
↷
✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

40. The graph of $y = f(x)$ and a table of values for an exponential function $y = g(x)$ are shown.



x	$g(x)$
-2	0.25
-1	0.5
0	1
1	2
2	4

Which statement correctly compares the graphs of the functions?

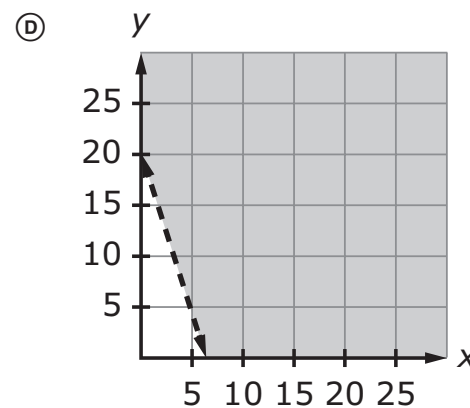
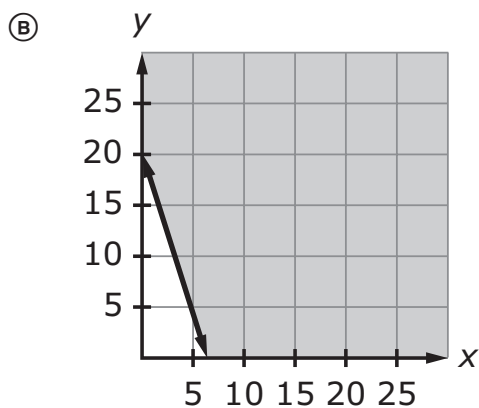
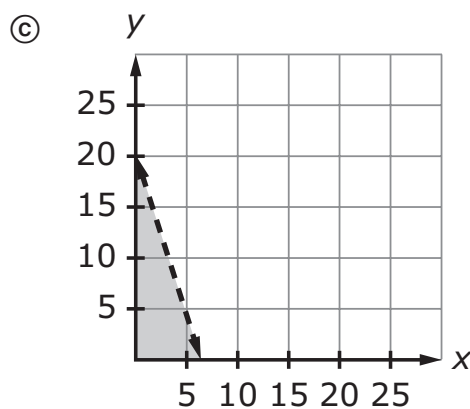
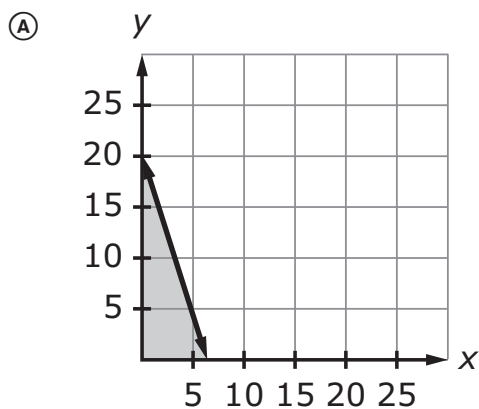
- Ⓐ As the values of x increase, $g(x)$ will eventually be greater than $f(x)$.
- Ⓑ As the values of x increase, $f(x)$ will eventually be greater than $g(x)$.
- Ⓒ The value of the y -intercept of $y = f(x)$ is greater than the value of the y -intercept of $y = g(x)$.
- Ⓓ The value of the y -intercept of $y = g(x)$ is greater than the value of the y -intercept of $y = f(x)$.

B.E.S.T. Algebra 1 EOC

- 41.** Sanjay buys x boxes of hot wings and y bottles of soda for a party. The numbers of boxes of hot wings and bottles of soda he can buy with \$40 can be represented with the inequality shown.

$$40 \geq 6x + 2y$$

Which graph represents all possible solutions for x and y ?



42. Janelle buys blackberries and raspberries to make pies.

- Blackberries cost \$8 per pound.
- Raspberries cost \$9 per pound.
- Janelle can spend a maximum of \$40.

Create an inequality to represent how many pounds of blackberries, b , and raspberries, r , Janelle can buy.

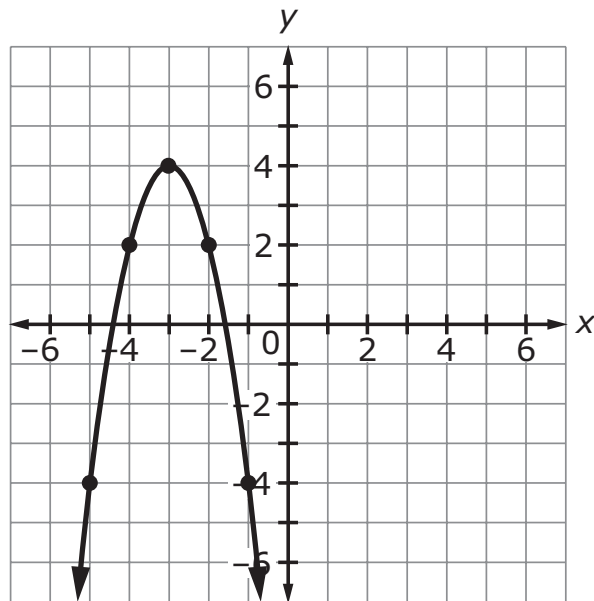
← → ↶ ↷ ✕											
1	2	3	b	r							
4	5	6	+	-	•	÷					
7	8	9	<	≤	=	≥	>				
	0		$\square^\square$	$\square_\square$	()		$\sqrt{\square}$	$\sqrt[\square]{\square}$	π	i	
.	-	$\frac{\square}{\square}$	sin	cos	tan	arcsin	arccos	arctan			

- 43.** Cynthia buys some tools. She creates exponential function V to predict the value of her tools after t years. She predicts her tools will have a value of \$380 after 1 year and a value of \$361 after 2 years.

Which equation represents function V ?

- Ⓐ $V(t) = 400(1 - 0.05)^t$
- Ⓑ $V(t) = 399(1 - 0.19)^t$
- Ⓒ $V(t) = 380(1 - 0.05)^t$
- Ⓓ $V(t) = 380(1 - 0.19)^t$

44. The graph of a quadratic function is shown.



Which equation represents the function?

- Ⓐ $y = -2(x + 3)^2 + 4$
- Ⓑ $y = -(x - 3)^2 + 4$
- Ⓒ $y = (x + 3)^2 + 4$
- Ⓓ $y = 6(x + 3)^2 - 4$

45. Select all the values that are equivalent to $\frac{\sqrt{24}}{\sqrt{8}}$.

Ⓐ $\sqrt{3}$

Ⓑ $\frac{1}{\sqrt{3}}$

Ⓒ $\sqrt{16}$

Ⓓ $\frac{2\sqrt{6}}{2\sqrt{2}}$

Ⓔ $\frac{4\sqrt{6}}{\sqrt{8}}$



THE B.E.S.T. STANDARDS

Benchmarks for Excellent Student Thinking

Algebra 1 EOC Assessment

Test Release: 2024

1. In baseball, a pitcher's earned run average is used to measure how effective the pitcher is. The earned run average, A , is calculated using the formula shown, where E is the number of earned runs and N is the number of innings the pitcher pitched.

$$A = \frac{9E}{N}$$

Which equation correctly solves the formula for the variable E ?

- Ⓐ $E = \frac{A}{9N}$
- Ⓑ $E = \frac{AN}{9}$
- Ⓒ $E = \frac{9A}{N}$
- Ⓓ $E = 9AN$

Answer Key: B

Percentage of Students Answering Correctly: 37%

Reporting Category: Expressions, Functions, and Data Analysis

Benchmark: MA.912.AR.1.2

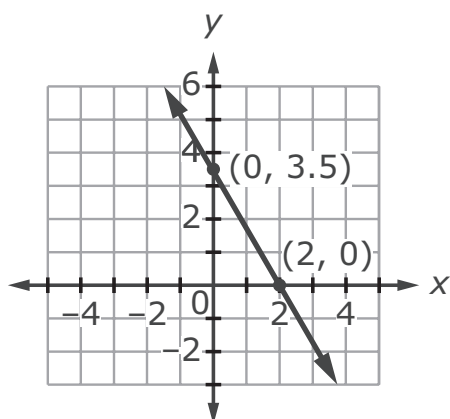
Benchmark Description: Rearrange equations or formulas to isolate a quantity of interest.

2. A function is shown.

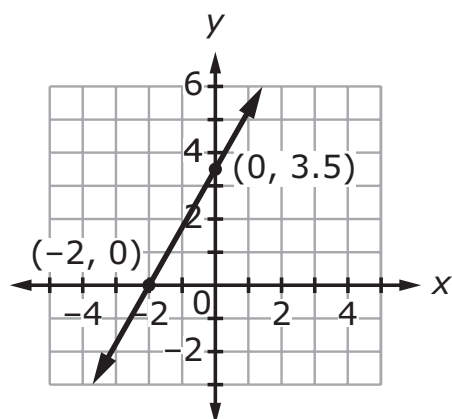
$$7x + 4y = 14$$

Which graph represents the function?

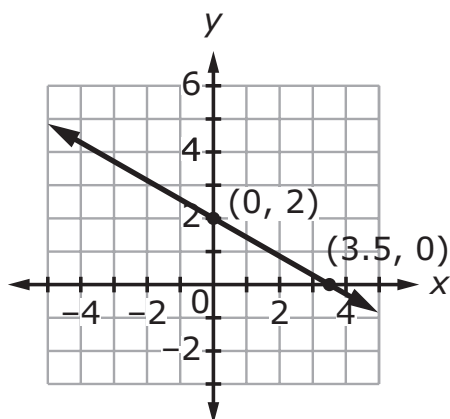
Ⓐ



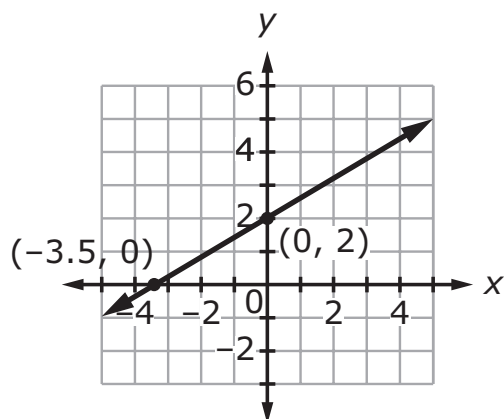
Ⓒ



Ⓑ



Ⓓ



Answer Key: A

Percentage of Students Answering Correctly: 49%

Reporting Category: Linear Relationships

Benchmark: MA.912.AR.2.4

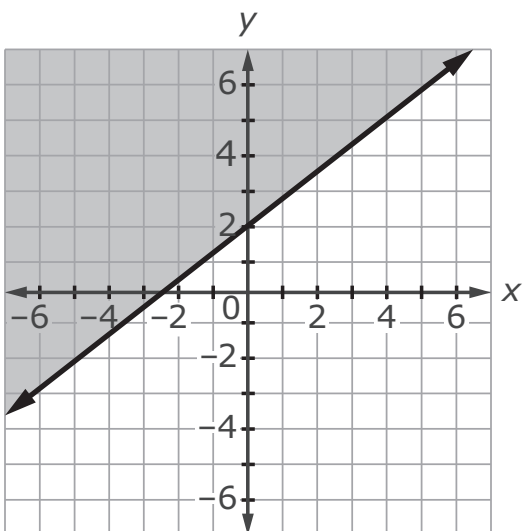
Benchmark Description: Given a table, equation or written description of a linear function, graph that function, and determine and interpret its key features.

3. An inequality is shown.

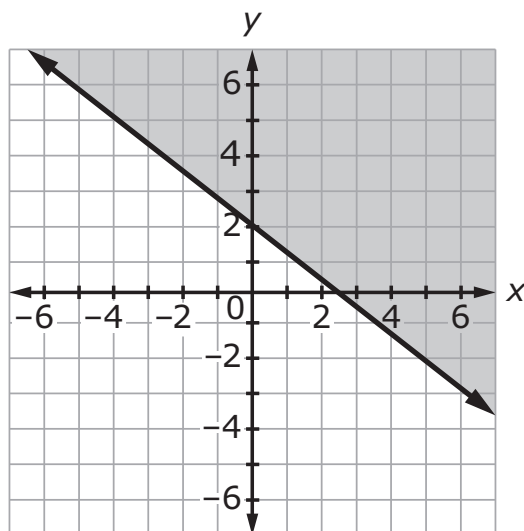
$$4x + 5y \geq 10$$

Which graph represents the solution set of the inequality?

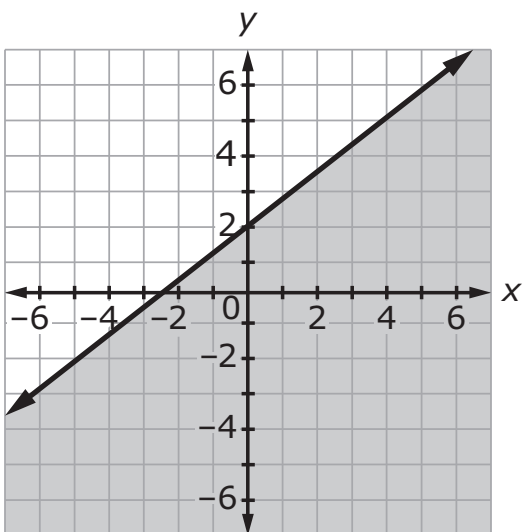
(A)



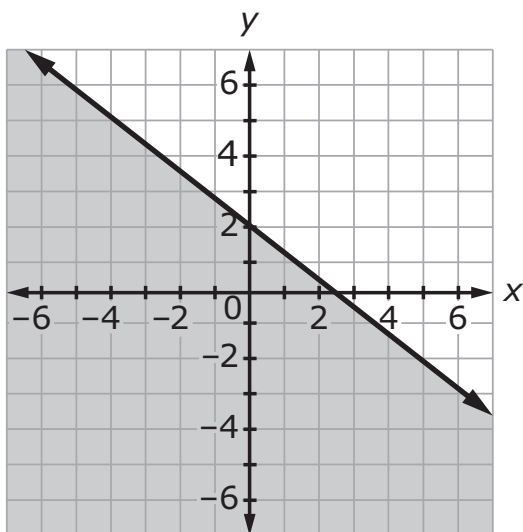
(C)



(B)



(D)



Answer Key: C

Percentage of Students Answering Correctly: 42%

Reporting Category: Linear Relationships

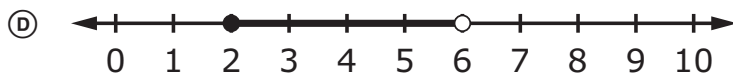
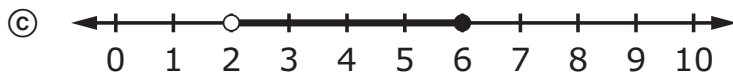
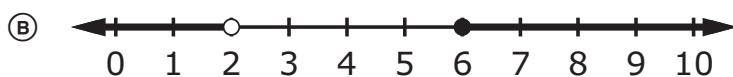
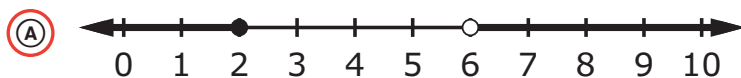
Benchmark: MA.912.AR.2.8

Benchmark Description: Given a mathematical or real-world context, graph the solution set to a two-variable linear inequality.

4. A compound inequality is shown.

$$2x - 1 \leq 3 \text{ or } \frac{1}{2}x + 4 > 7$$

Which graph represents the solution set of the compound inequality?



Answer Key: A

Percentage of Students Answering Correctly: 51%

Reporting Category: Linear Relationships

Benchmark: MA.912.AR.2.6

Benchmark Description: Given a mathematical or real-world context, write and solve one-variable linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.

5. An inequality is shown.

$$-3x + 7 \geq -x - 5$$

What is the solution set for x ?

- Ⓐ $\{x \mid x \leq 6\}$
- Ⓑ $\{x \mid x \geq 6\}$
- Ⓒ $\{x \mid x \leq -6\}$
- Ⓓ $\{x \mid x \geq -6\}$

Answer Key: A

Percentage of Students Answering Correctly: 38%

Reporting Category: Linear Relationships

Benchmark: MA.912.AR.2.6

Benchmark Description: Given a mathematical or real-world context, write and solve one-variable linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.

6. What is the sum of $(2x^3 + 4x)$ and $(3x^3 + 5x)$?

- Ⓐ $14x^6$
- Ⓑ $14x^8$
- Ⓒ $5x^3 + 9x$
- Ⓓ $5x^6 + 9x^2$

Answer Key: C

Percentage of Students Answering Correctly: 49%

Reporting Category: Non-Linear Relationships

Benchmark: MA.912.AR.1.3

Benchmark Description: Add, subtract and multiply polynomial expressions with rational number coefficients.

7. A system of equations is shown.

$$\begin{aligned} y &= 2x + 2 \\ 2x + 5y &= 19 \end{aligned}$$

What is the solution to the system?

Ⓐ $\left(\frac{3}{4}, 3\frac{1}{2}\right)$

Ⓑ $\left(1\frac{1}{8}, 4\frac{1}{4}\right)$

Ⓒ $\left(1\frac{2}{3}, 5\frac{1}{3}\right)$

Ⓓ $\left(3\frac{1}{4}, 2\frac{1}{2}\right)$

Answer Key: A

Percentage of Students Answering Correctly: 41%

Reporting Category: Linear Relationships

Benchmark: MA.912.AR.9.1

Benchmark Description: Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.

8. Which expression is equivalent to $\sqrt{3}$?

Ⓐ $2^{\frac{1}{3}}$

Ⓑ $3^{\frac{1}{2}}$

Ⓒ $\left(\frac{1}{2}\right)^3$

Ⓓ $\left(\frac{1}{3}\right)^2$

Answer Key: B

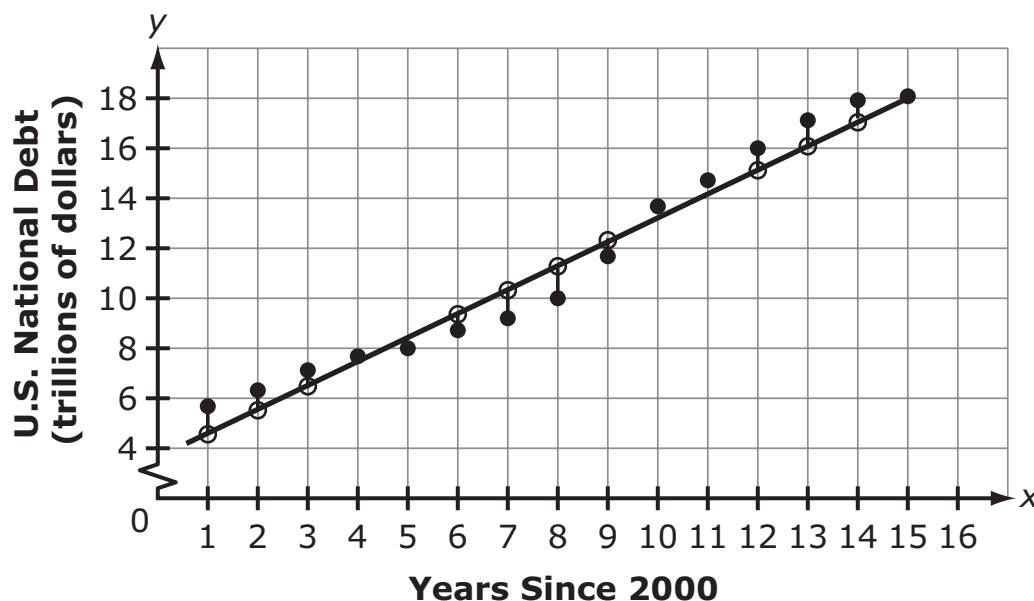
Percentage of Students Answering Correctly: 45%

Reporting Category: Expressions, Functions, and Data Analysis

Benchmark: MA.912.NSO.1.1

Benchmark Description: Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

9. A scatter plot of the U.S. national debt from 2001 to 2016, including a linear equation and its residuals modeling the data, is shown.



Which statement correctly interprets the scatter plot?

- (A) The scatter plot shows a weak positive correlation.
- (B) The scatter plot shows a weak negative correlation.
- (C) The scatter plot shows a strong positive correlation.
- (D) The scatter plot shows a strong negative correlation.

Answer Key: C

Percentage of Students Answering Correctly: 59%

Reporting Category: Linear Relationships

Benchmark: MA.912.DP.2.6

Benchmark Description: Given a scatter plot with a line of fit and residuals, determine the strength and direction of the correlation. Interpret strength and direction within a real-world context.

10. An equation is shown.

$$3x^2 - 21 = 222$$

What is one exact solution to the equation?

$x =$

← → ↶ ↷ ✖											
1	2	3	+	-	•	÷					
4	5	6	<	≤	=	≥	>				
7	8	9	$\square^\square$	$\square_\square$	()		$\sqrt{\square}$	$\sqrt[\square]{\square}$	π	i	
	0		sin	cos	tan	arcsin	arccos	arctan			
.	-	$\frac{\square}{\square}$									

Answer Key: 9, -9, or any equivalent values

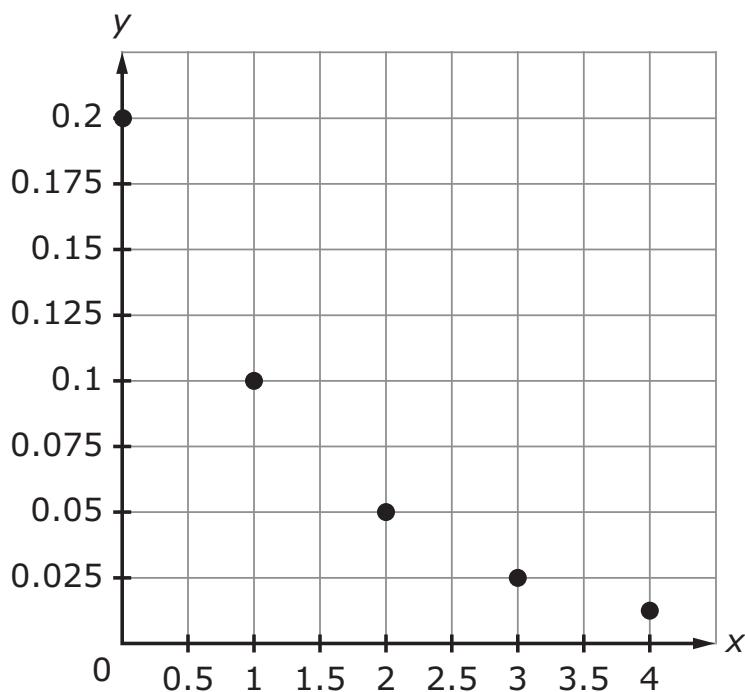
Percentage of Students Answering Correctly: 34%

Reporting Category: Non-Linear Relationships

Benchmark: MA.912.AR.3.1

Benchmark Description: Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

11. Five points from a function, f , are plotted in the first quadrant of the coordinate plane, as shown.



Which equation could represent the function?

- Ⓐ $f(x) = 0.2(1 - 0.1)^x$
- Ⓑ $f(x) = 0.2(1 - 0.5)^x$
- Ⓒ $f(x) = 0.1(1 - 0.05)^x$
- Ⓓ $f(x) = 0.1(1 - 0.25)^x$

Answer Key: B

Percentage of Students Answering Correctly: 43%

Reporting Category: Non-Linear Relationships

Benchmark: MA.912.AR.5.4

Benchmark Description: Write an exponential function to represent a relationship between two quantities from a graph, a written description or a table of values within a mathematical or real-world context.

12. Which expression is equivalent to $(7k + 2)(5k - 9)$?

- Ⓐ $35k^2 - 18$
- Ⓑ $-18k - 18$
- Ⓒ $35k^2 - 53k - 18$
- Ⓓ $35k^2 - 73k - 18$

Answer Key: C

Percentage of Students Answering Correctly: 52%

Reporting Category: Non-Linear Relationships

Benchmark: MA.912.AR.1.3

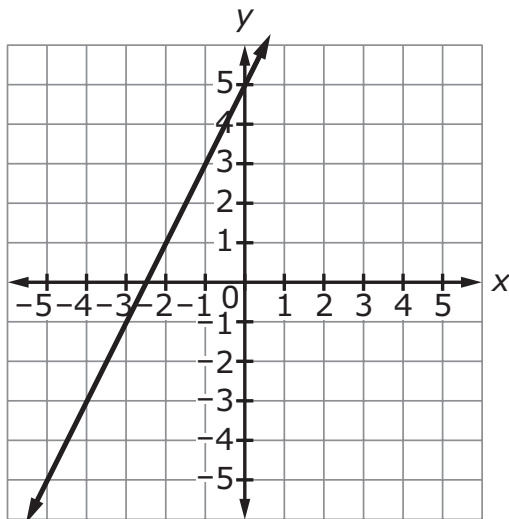
Benchmark Description: Add, subtract and multiply polynomial expressions with rational number coefficients.

13. A function is shown.

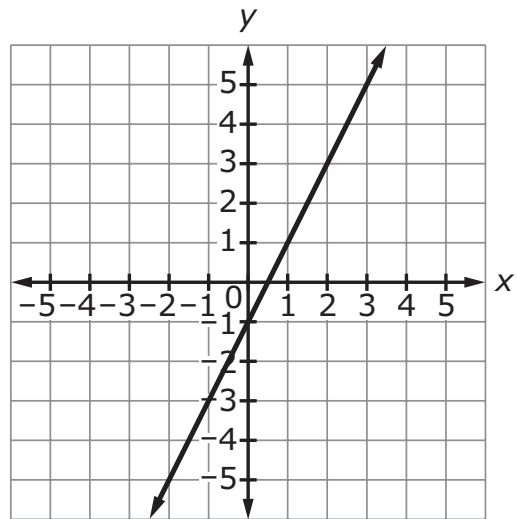
$$y + 1 = 2(x + 3)$$

Which graph represents the function?

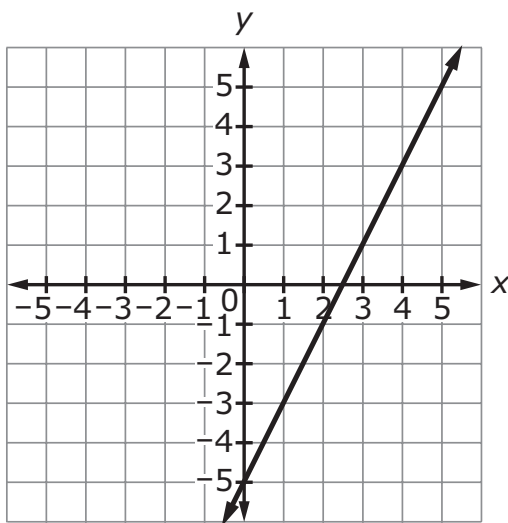
Ⓐ



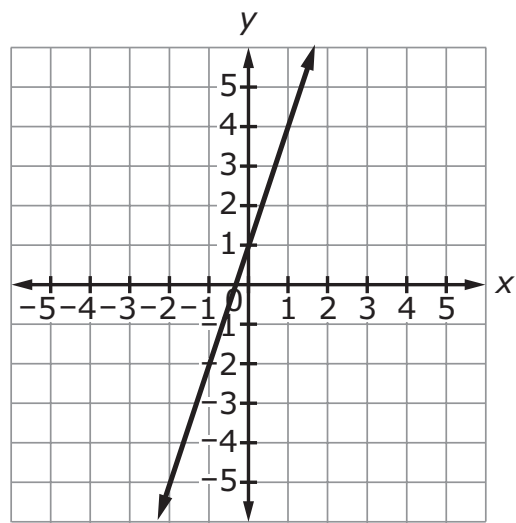
Ⓒ



Ⓑ



Ⓓ



Florida Test Release Support Document

Answer Key: A

Percentage of Students Answering Correctly: 36%

Reporting Category: Linear Relationships

Benchmark: MA.912.AR.2.4

Benchmark Description: Given a table, equation or written description of a linear function, graph that function, and determine and interpret its key features.

- 14.** Garrett buys a large bag of dog food. He feeds his dog 2 cups of dog food each day.

Which statement describes the type of function that represents the amount of dog food in the bag after x days?

- Ⓐ positive linear
- Ⓑ negative linear
- Ⓒ exponential
- Ⓓ quadratic

Answer Key: B

Percentage of Students Answering Correctly: 53%

Reporting Category: Expressions, Functions, and Data Analysis

Benchmark: MA.912.F.1.8

Benchmark Description: Determine whether a linear, quadratic or exponential function best models a given real-world situation.

15. Which table could represent a quadratic function?

(A)

x	$f(x)$
0	3
1	9
2	27
3	81

(C)

x	$h(x)$
-1	6
0	5
1	6
2	9

(B)

x	$g(x)$
0	-2
1	-4
2	-6
3	-8

(D)

x	$j(x)$
-3	7
-2	7
-1	7
0	7

Answer Key: C

Percentage of Students Answering Correctly: 34%

Reporting Category: Expressions, Functions, and Data Analysis

Benchmark: MA.912.F.1.1

Benchmark Description: Given an equation or graph that defines a function, classify the function type. Given an input output table, determine a function type that could represent it.

16. Which function has the greatest slope?

Ⓐ $y = \frac{5}{3}x + 1$

Ⓑ $y = -x + 8$

Ⓒ $y = \frac{3}{4}x - \frac{5}{4}$

Ⓓ $y = \frac{3}{2}x + 3$

Answer Key: A

Percentage of Students Answering Correctly: 49%

Reporting Category: Linear Relationships

Benchmark: MA.912.F.1.5

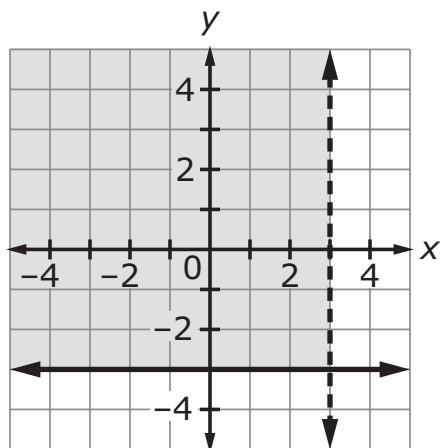
Benchmark Description: Compare key features of linear functions each represented algebraically, graphically, in tables or written descriptions.

17. A system of inequalities is shown.

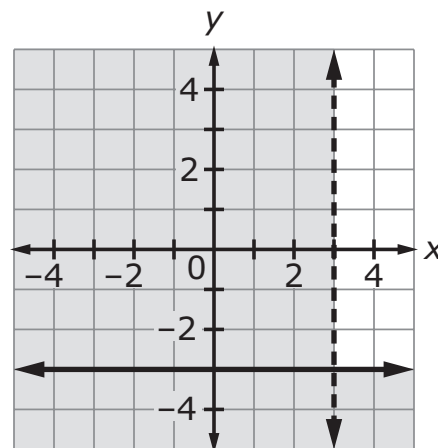
$$\begin{aligned} x &< 3 \\ y &\geq -3 \end{aligned}$$

Which graph represents the solution set?

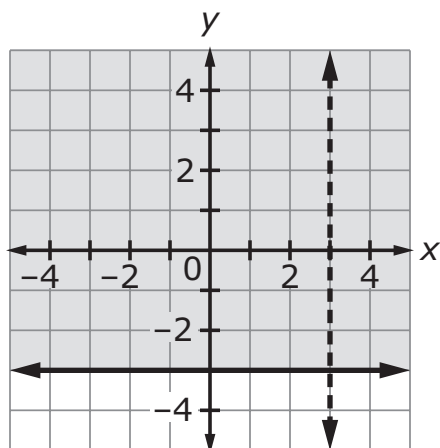
Ⓐ



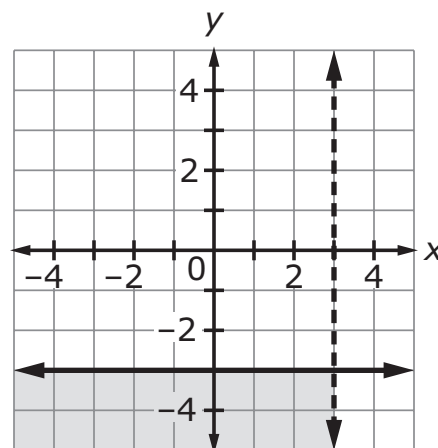
Ⓒ



Ⓑ



Ⓓ



Answer Key: A

Percentage of Students Answering Correctly: 59%

Reporting Category: Linear Relationships

Benchmark: MA.912.AR.9.4

Benchmark Description: Graph the solution set of a system of two-variable linear inequalities.

- 18.** A ball was dropped from a balcony at a height of 20 feet above level ground. The height, in feet, of the ball after t seconds is modeled by the function shown.

$$y = -16t^2 + 20$$

What is the range of the function in terms of the situation described?

- Ⓐ $0 \leq t$
- Ⓑ $0 \leq y$
- Ⓒ $0 \leq t \leq 20$
- Ⓓ $0 \leq y \leq 20$

Answer Key: D

Percentage of Students Answering Correctly: 31%

Reporting Category: Non-Linear Relationships

Benchmark: MA.912.AR.3.8

Benchmark Description: Solve and graph mathematical and real-world problems that are modeled with quadratic functions. Interpret key features and determine domain constraints in terms of the context.

19. Which equation models exponential growth?

Ⓐ $y = 2\left(\frac{1}{5}\right)^x$

Ⓑ $y = 2(5.0)^x$

Ⓒ $y = 2\left(1 - \frac{1}{5}\right)^x$

Ⓓ $y = 2(1 - 0.5)^x$

Answer Key: B

Percentage of Students Answering Correctly: 58%

Reporting Category: Non-Linear Relationships

Benchmark: MA.912.AR.5.3

Benchmark Description: Given a mathematical or real-world context, classify an exponential function as representing growth or decay.

20. Which function decays exponentially?

Ⓐ $f(x) = \frac{1}{5}x$

Ⓑ $g(x) = 5x$

Ⓒ $h(x) = 5^x$

Ⓓ $j(x) = \left(\frac{1}{5}\right)^x$

Answer Key: D

Percentage of Students Answering Correctly: 47%

Reporting Category: Non-Linear Relationships

Benchmark: MA.912.AR.5.3

Benchmark Description: Given a mathematical or real-world context, classify an exponential function as representing growth or decay.

21. A line with a slope of -0.2 passes through the point $(11, 5)$.

Select numbers to complete the equation of the line in point-slope form.

$$y - [\textcircled{A} -0.2 \textcircled{B} 0 \textcircled{C} 5 \textcircled{D} 11] = [\textcircled{A} -0.2 \textcircled{B} 0 \textcircled{C} 5 \textcircled{D} 11]$$

$$(x - [\textcircled{A} -0.2 \textcircled{B} 0 \textcircled{C} 5 \textcircled{D} 11])$$

Answer Key: C, A, D

Percentage of Students Answering Correctly: 46%

Reporting Category: Linear Relationships

Benchmark: MA.912.AR.2.2

Benchmark Description: Write a linear two-variable equation to represent relationships between quantities from a graph, a written description or a table of values within a mathematical or real-world context.

22. An expression is given, where $x \neq 0$.

$$\frac{4x^2 - 18x + 6}{2x}$$

Which expression is equivalent?

- Ⓐ $2x - 9 + \frac{3}{x}$
- Ⓑ $2x^2 - 16x + 4$
- Ⓒ $2x^3 - 9x^2 + 3x$
- Ⓓ $\frac{2}{x^3} - \frac{9}{x^2} + \frac{3}{x}$

Answer Key: A

Percentage of Students Answering Correctly: 45%

Reporting Category: Non-Linear Relationships

Benchmark: MA.912.AR.1.4

Benchmark Description: Divide a polynomial expression by a monomial expression with rational number coefficients.

23. An equation is given.

$$y = (x - 3)^2 + 1$$

Which type of function does the equation represent?

- Ⓐ cubic
- Ⓑ quadratic
- Ⓒ exponential
- Ⓓ square root

Answer Key: B

Percentage of Students Answering Correctly: 51%

Reporting Category: Expressions, Functions, and Data Analysis

Benchmark: MA.912.F.1.1

Benchmark Description: Given an equation or graph that defines a function, classify the function type. Given an input output table, determine a function type that could represent it.

24. A function is given

$$h(x) = \frac{1}{5}x^2$$

How can be $h(15)$ evaluated?

- Ⓐ by multiplying $\frac{1}{5}$ by the square of 15
- Ⓑ by multiplying the square of $\frac{1}{5}$ by the square of 15
- Ⓒ by multiplying $\frac{1}{5}$ by the square of 15, then dividing by 15
- Ⓓ by multiplying the square of $\frac{1}{5}$ by the square of 15, then dividing by 15

Answer Key: A

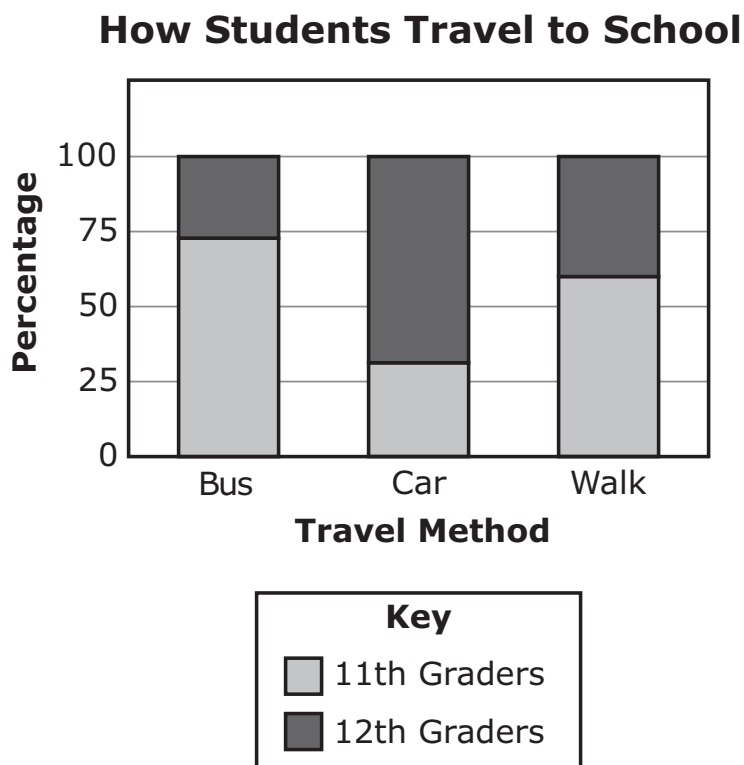
Percentage of Students Answering Correctly: 48%

Reporting Category: Expressions, Functions, and Data Analysis

Benchmark: MA.912.F.1.2

Benchmark Description: Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output.

25. Maribel asked a random group of upper-class students at her school which grade they are in and how they traveled to school in the morning. The data are shown in the segmented bar chart.



Which pair of terms describes the data?

- (A) bivariate and numerical
- (B) bivariate and categorical
- (C) univariate and numerical
- (D) univariate and categorical

Florida Test Release Support Document

Answer Key: B

Percentage of Students Answering Correctly: 52%

Reporting Category: Expressions, Functions, and Data Analysis

Benchmark: MA.912.DP.1.2

Benchmark Description: Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate and interpret the different components and quantities in the display.

26. An expression is given.

$$42r^3s^2 - 35r^2s^3$$

Which expression is equivalent?

- Ⓐ $7rs(6s - 5r)$
- Ⓑ $7rs(6rs - 5rs)$
- Ⓒ $7r^2s^2(6r - 5s)$
- Ⓓ $7r^2s^2(6s - 5r)$

Answer Key: C

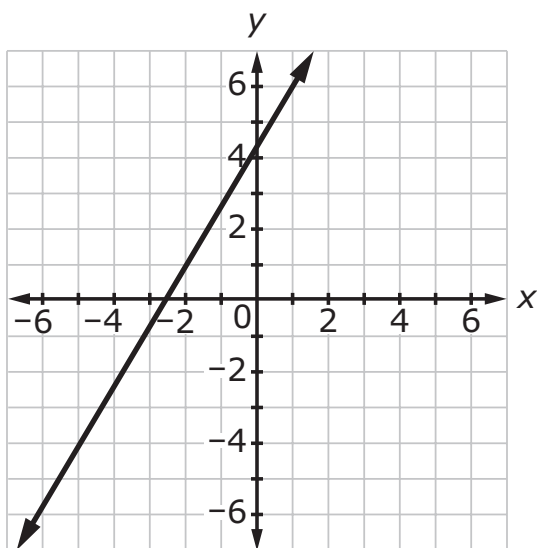
Percentage of Students Answering Correctly: 60%

Reporting Category: Non-Linear Relationships

Benchmark: MA.912.AR.1.7

Benchmark Description: Rewrite a polynomial expression as a product of polynomials over the real number system.

27. The graph of a line is shown.



Which equation represents the line?

Ⓐ $y = \frac{3}{5}x - \frac{13}{5}$

Ⓑ $y = \frac{3}{5}x + \frac{13}{3}$

Ⓒ $y = \frac{5}{3}x - \frac{13}{5}$

Ⓓ $y = \frac{5}{3}x + \frac{13}{3}$

Answer Key: D

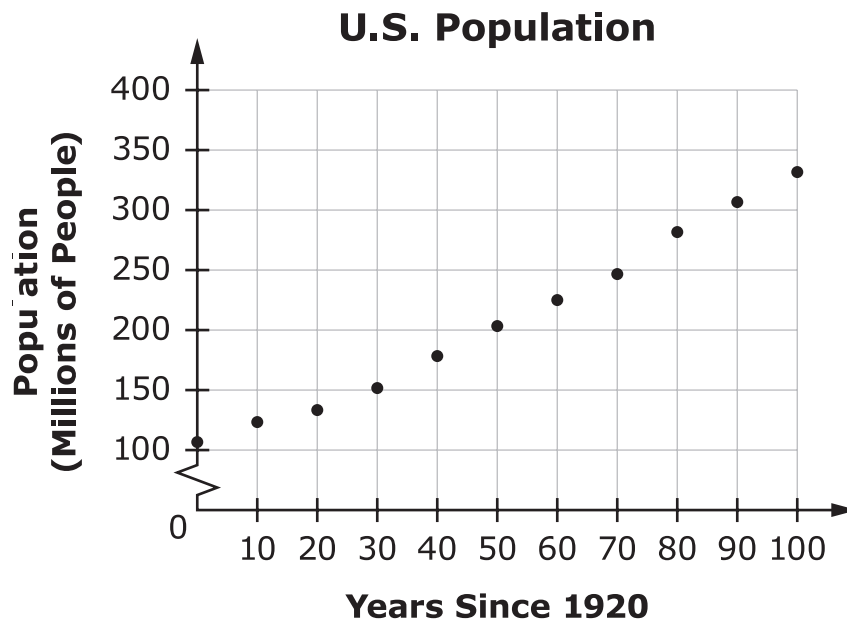
Percentage of Students Answering Correctly: 36%

Reporting Category: Linear Relationships

Benchmark: MA.912.AR.2.2

Benchmark Description: Write a linear two-variable equation to represent relationships between quantities from a graph, a written description or a table of values within a mathematical or real-world context.

28. Darren collects data on the U.S. population from 1920 to 2020. He graphs the population and the number of years since 1920 in the scatter plot shown.



He determines the equation for the line of best fit to be $y = 2.33x + 92.05$ where y represents the population, in millions of people, and x represents the number of years since 1920.

Based on Darren's model, what is the projected U.S. population, in millions of people, for 2040?

371.65

←
→
↶
↷
✖

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Answer Key: 371.65, 372, or any equivalent values

Percentage of Students Answering Correctly: 27%

Reporting Category: Linear Relationships

Benchmark: MA.912.DP.2.4

Benchmark Description: Fit a linear function to bivariate numerical data that suggests a linear association and interpret the slope and y-intercept of the model. Use the model to solve real-world problems in terms of the context of the data.

Florida Test Release Support Document

29. Bashir has \$3410 to invest in two different investment accounts, Account A and Account B. The accounts earn simple interest for one year.

- Account A earns 6% interest.
- Account B earns 2% interest.
- He wants to earn at least \$196 in interest.

Bashir puts x dollars in Account A and y dollars in Account B.

Which system of constraints represents the situation?

- ☐ Ⓐ $x + y = 3410$
 $0.02x + 0.06y \geq 196$
- ☒ Ⓑ $x + y = 3410$
 $0.06x + 0.02y \geq 196$
- ☐ Ⓒ $0.02x + 0.06y = 3410$
 $x + y \geq 196$
- ☐ Ⓓ $0.06x + 0.02y \geq 3410$
 $x + y \geq 196$

Answer Key: B

Percentage of Students Answering Correctly: 62%

Reporting Category: Linear Relationships

Benchmark: MA.912.AR.9.6

Benchmark Description: Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.

- 30.** A rental company charges \$2 per hour to rent a bicycle. A table showing the prices of renting one bicycle for different amounts of time is shown.

Time (hours)	Price (dollars)
1	2
2	4
3	6

Which type of function models the price for renting one bicycle as a function of time?

- ☒ (A) increasing linear
- ☐ (B) decreasing linear
- ☐ (C) increasing exponential
- ☐ (D) decreasing exponential

Answer Key: A

Percentage of Students Answering Correctly: 76%

Reporting Category: Expressions, Functions, and Data Analysis

Benchmark: MA.912.F.1.8

Benchmark Description: Determine whether a linear, quadratic or exponential function best models a given real-world situation.

- 31.** An artist earns a profit from selling y paintings. His profit, P , in dollars, is modeled with the given equation.

$$P = (200y + 23.25) - (10.53y)$$

His profit is the difference in the amount of money he earns from selling his paintings and the cost of making the paintings.

Which expression represents the cost of making y paintings?

- Ⓐ 23.25
- Ⓑ 10.53y
- Ⓒ $200y + 23.25$
- Ⓓ $189.47y + 23.25$

Answer Key: B

Percentage of Students Answering Correctly: 39%

Reporting Category: Expressions, Functions, and Data Analysis

Benchmark: MA.912.AR.1.1

Benchmark Description: Identify and interpret parts of an equation or expression that represent a quantity in terms of a mathematical or real-world context, including viewing one or more of their parts as a single entity.

- 32.** Cardiac output, D , is the amount of blood that the heart pumps per minute. The value of D can be found using the formula shown, where V is the amount of blood that the heart pumps with each heartbeat and R is the number of heartbeats per minute.

$$D = V \cdot R$$

Complete the equation to represent V in terms of D and R .

$$V = \frac{D}{R}$$

Answer Key: $\frac{D}{R}$, or any equivalent expression

Percentage of Students Answering Correctly: 36%

Reporting Category: Expressions, Functions, and Data Analysis

Benchmark: MA.912.AR.1.2

Benchmark Description: Rearrange equations or formulas to isolate a quantity of interest.

Florida Test Release Support Document

- 33.** Two divers are underwater. One diver is at a depth of 32 meters, while the other diver is at a different depth. The given equation can be used to find the possible depths, x , in meters, of the other diver.

$$|x-32| = 11$$

Enter values to complete the sentence about the other diver's depth, in meters.

The other diver is at a depth of either

21

←

→

↶

↷

✕

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

or

43

←

→

↶

↷

✕

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

meters.

Answer Key: The other diver is at a depth of either 21 or 43 meters, or any equivalent values. 21 and 43 may be switched.

Percentage of Students Answering Correctly: 27%

Reporting Category: Non-Linear Relationships

Benchmark: MA.912.AR.4.1

Benchmark Description: Given a mathematical or real-world context, write and solve one-variable absolute value equations.

34. The equation of a linear function, q , and a table of values of a linear function, r , are shown.

$$q(x) = 2x - 1$$

x	$r(x)$
0	-2
1	4
2	10
3	16

Which statement correctly compares the slopes and y -intercepts of the two functions?

- Ⓐ Function r has a lesser y -intercept and a slope that is 2 times the slope of function q .
- Ⓑ Function r has a lesser y -intercept and a slope that is 3 times the slope of function q .
- Ⓒ Function r has a greater y -intercept and a slope that is 2 times the slope of function q .
- Ⓓ Function r has a greater y -intercept and a slope that is 3 times the slope of function q .

Answer Key: B

Percentage of Students Answering Correctly: 31%

Reporting Category: Linear Relationships

Benchmark: MA.912.F.1.5

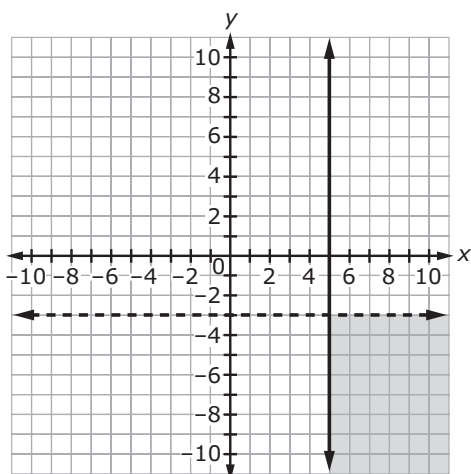
Benchmark Description: Compare key features of linear functions each represented algebraically, graphically, in tables or written descriptions.

35. A system of inequalities is given.

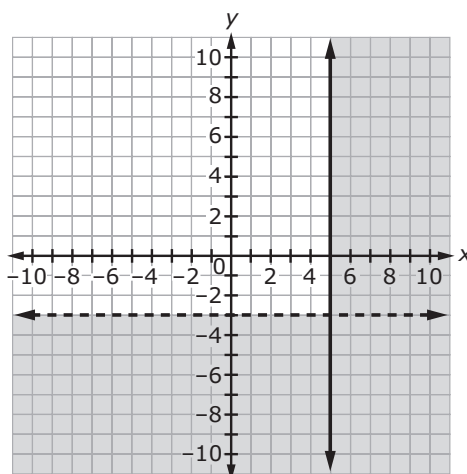
$$\begin{aligned} x &\geq 5 \\ y &< -3 \end{aligned}$$

Which graph represents the solution set?

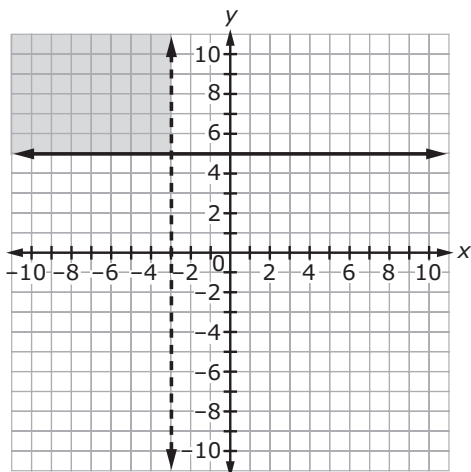
(A)



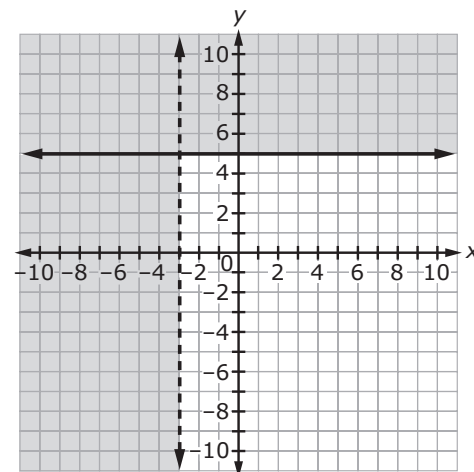
(C)



(B)



(D)



Answer Key: A

Percentage of Students Answering Correctly: 41%

Reporting Category: Linear Relationships

Benchmark: MA.912.AR.9.4

Benchmark Description: Graph the solution set of a system of two-variable linear inequalities.

36. An equation is shown.

$$x^2 - 15x = -14$$

What are the two solutions to the equation?

$x =$

1

←
→
↶
↷
✕

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

$x =$

14

←
→
↶
↷
✕

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Answer Key: 1 and 14, or any equivalent value

Percentage of Students Answering Correctly: 19%

Reporting Category: Non-Linear Relationships

Benchmark: MA.912.AR.3.1

Benchmark Description: Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

37. A function, f , is given.

$$f(x) = x^2 + 12x + 20$$

Select terms to describe the end behavior of the function.

As x approaches $-\infty$, $f(x)$ approaches [☐ A $-\infty$ ☒ B ∞ ☐ C 0] .

As x approaches ∞ , $f(x)$ approaches [☐ A $-\infty$ ☒ B ∞ ☐ C 0] .

Answer Key: B, B

Percentage of Students Answering Correctly: 12%

Reporting Category: Non-Linear Relationships

Benchmark: MA.912.AR.3.7

Benchmark Description: Given a table, equation or written description of a quadratic function, graph that function, and determine and interpret its key features.

38. The expression $\sqrt[3]{18} \cdot 5\sqrt[3]{6}$ is equivalent to $15\sqrt[3]{x}$.

What is the value of x ?

4

← → ↶ ↷ ✕

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\square}{\square}$

Answer Key: 4, or any equivalent value

Percentage of Students Answering Correctly: 16%

Reporting Category: Expressions, Functions, and Data Analysis

Benchmark: MA.912.NSO.1.4

Benchmark Description: Apply previous understanding of operations with rational numbers to add, subtract, multiply and divide numerical radicals.

- 39.** Margo surveys 150 high school juniors and seniors about their favorite types of dipping sauce. She records the results in the table shown.

Favorite Dipping Sauce

	Barbeque	Honey Mustard	Ranch	Sweet and Sour
Juniors	19	26	23	11
Seniors	7	22	30	12

How many more high school juniors and seniors chose honey mustard sauce than sweet and sour sauce?

25

←
→
↶
↷
✕

1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Answer Key: 25, or any equivalent value

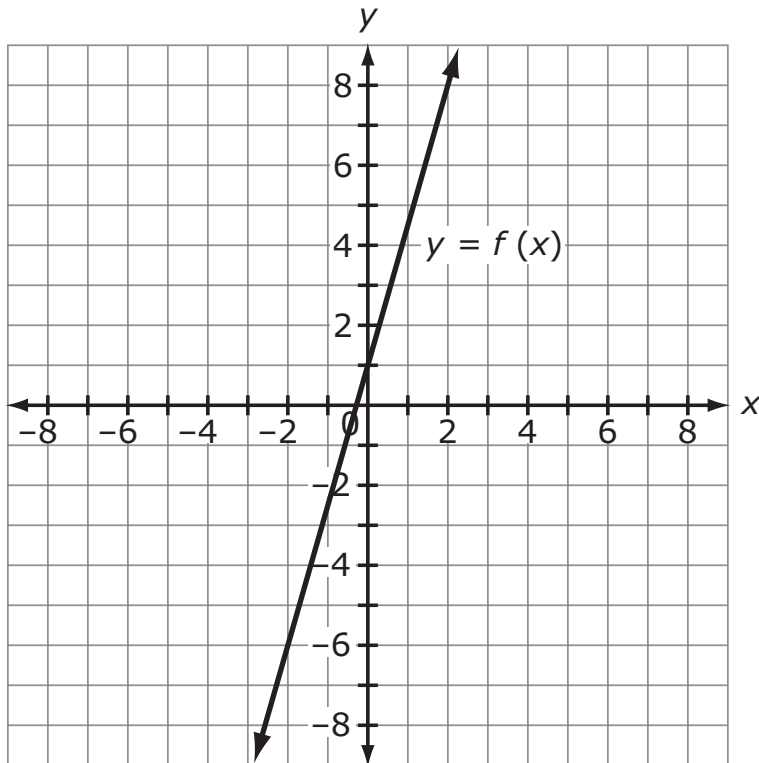
Percentage of Students Answering Correctly: 68%

Reporting Category: Expressions, Functions, and Data Analysis

Benchmark: MA.912.DP.3.1

Benchmark Description: Construct a two-way frequency table summarizing bivariate categorical data. Interpret joint and marginal frequencies and determine possible associations in terms of a real-world context.

40. The graph of $y = f(x)$ and a table of values for an exponential function $y = g(x)$ are shown.



x	$g(x)$
-2	0.25
-1	0.5
0	1
1	2
2	4

Which statement correctly compares the graphs of the functions?

- Ⓐ As the values of x increase, $g(x)$ will eventually be greater than $f(x)$.
- Ⓑ As the values of x increase, $f(x)$ will eventually be greater than $g(x)$.
- Ⓒ The value of the y -intercept of $y = f(x)$ is greater than the value of the y -intercept of $y = g(x)$.
- Ⓓ The value of the y -intercept of $y = g(x)$ is greater than the value of the y -intercept of $y = f(x)$.

Answer Key: A

Percentage of Students Answering Correctly: 40%

Reporting Category: Expressions, Functions, and Data Analysis

Benchmark: MA.912.F.1.6

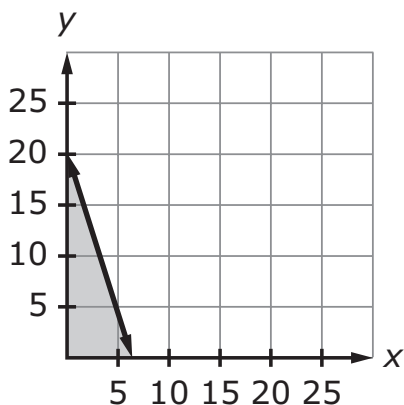
Benchmark Description: Compare key features of linear and nonlinear functions each represented algebraically, graphically, in tables or written descriptions.

- 41.** Sanjay buys x boxes of hot wings and y bottles of soda for a party. The numbers of boxes of hot wings and bottles of soda he can buy with \$40 can be represented with the inequality shown.

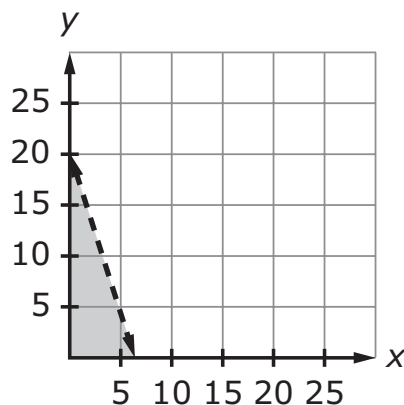
$$40 \geq 6x + 2y$$

Which graph represents all possible solutions for x and y ?

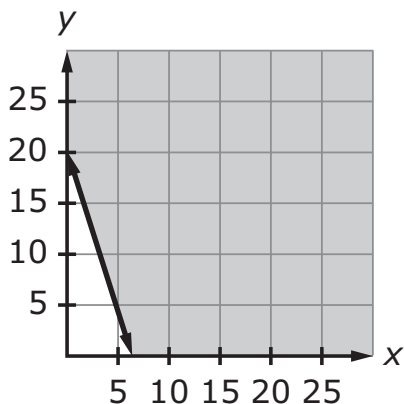
Ⓐ



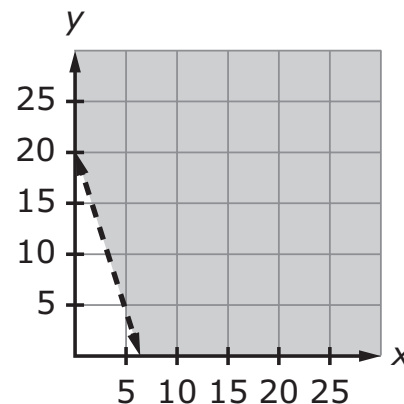
Ⓒ



Ⓑ



Ⓓ



Answer Key: A

Percentage of Students Answering Correctly: 32%

Reporting Category: Linear Relationships

Benchmark: MA.912.AR.2.8

Benchmark Description: Given a mathematical or real-world context, graph the solution set to a two-variable linear inequality.

42. Janelle buys blackberries and raspberries to make pies.

- Blackberries cost \$8 per pound.
- Raspberries cost \$9 per pound.
- Janelle can spend a maximum of \$40.

Create an inequality to represent how many pounds of blackberries, b , and raspberries, r , Janelle can buy.

$8b + 9r \leq 40$

←
→
↶
↷
✖

1	2	3	b	r															
4	5	6	+	-	•	÷													
7	8	9	<	≤	=	≥	>												
	0		$\square^\square$	$\square_\square$	()		$\sqrt{\square}$	$\sqrt[\square]{\square}$	π	i									
.	-	$\frac{\square}{\square}$	sin	cos	tan	arcsin	arccos	arctan											

Answer Key: $8b + 9r \leq 40$, or any equivalent inequality

Percentage of Students Answering Correctly: 25%

Reporting Category: Linear Relationships

Benchmark: MA.912.AR.2.7

Benchmark Description: Write two-variable linear inequalities to represent relationships between quantities from a graph or a written description within a mathematical or real-world context.

- 43.** Cynthia buys some tools. She creates exponential function V to predict the value of her tools after t years. She predicts her tools will have a value of \$380 after 1 year and a value of \$361 after 2 years.

Which equation represents function V ?

- Ⓐ $V(t) = 400(1 - 0.05)^t$
- Ⓑ $V(t) = 399(1 - 0.19)^t$
- Ⓒ $V(t) = 380(1 - 0.05)^t$
- Ⓓ $V(t) = 380(1 - 0.19)^t$

Answer Key: A

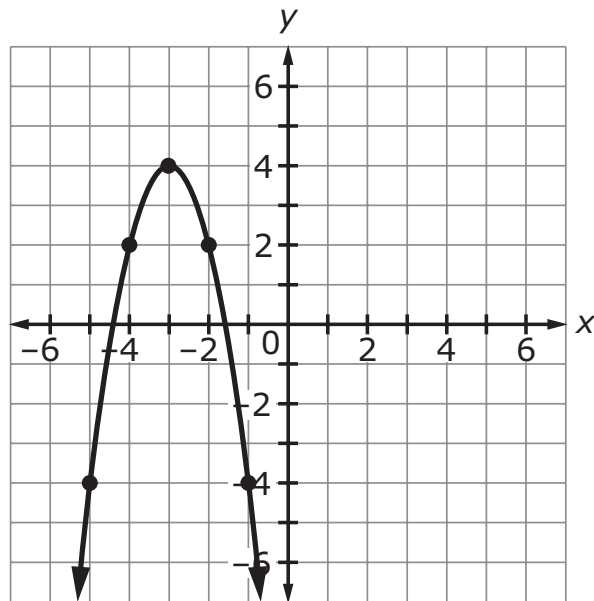
Percentage of Students Answering Correctly: 14%

Reporting Category: Non-Linear Relationships

Benchmark: MA.912.AR.5.4

Benchmark Description: Write an exponential function to represent a relationship between two quantities from a graph, a written description or a table of values within a mathematical or real-world context.

44. The graph of a quadratic function is shown.



Which equation represents the function?

- Ⓐ $y = -2(x + 3)^2 + 4$
- Ⓑ $y = -(x - 3)^2 + 4$
- Ⓒ $y = (x + 3)^2 + 4$
- Ⓓ $y = 6(x + 3)^2 - 4$

Answer Key: A

Percentage of Students Answering Correctly: 42%

Reporting Category: Non-Linear Relationships

Benchmark: MA.912.AR.3.4

Benchmark Description: Write a quadratic function to represent the relationship between two quantities from a graph, a written description or a table of values within a mathematical or real-world context.

45. Select all the values that are equivalent to $\frac{\sqrt{24}}{\sqrt{8}}$.

☒ (A) $\sqrt{3}$

☐ (B) $\frac{1}{\sqrt{3}}$

☐ (C) $\sqrt{16}$

☒ (D) $\frac{2\sqrt{6}}{2\sqrt{2}}$

☐ (E) $\frac{4\sqrt{6}}{\sqrt{8}}$

Answer Key: A, D

Percentage of Students Answering Correctly: 15%

Reporting Category: Expressions, Functions, and Data Analysis

Benchmark: MA.912.NSO.1.4

Benchmark Description: Apply previous understanding of operations with rational numbers to add, subtract, multiply and divide numerical radicals.